

Zachary K. Kingston

Contact Information

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Research Interests

I am interested in fast, reactive planning for robots over large time scales satisfying complex constraints. I also want these methods to generalize to any system (e.g., manipulators, soft robots, mobile robots, etc.) and provide guarantees on performance, completeness, or solution quality. I am also interested in the intersection between theory and implementation, where software engineering and hardware acceleration (e.g., SIMD, CUDA, etc.) can synergize to bring real-time performance to even heavily constrained platforms.

Professional Experience

Purdue University <i>at West Lafayette, IN</i>	Assistant Professor <i>in the Department of Computer Science</i>	Aug. 2024–Present
Kavraki Lab <i>at Rice University, Houston, TX</i>	Post-Doctoral Researcher and Lab Manager <i>Supervisor: Dr. Lydia Kavraki</i>	Dec. 2021–Aug. 2024 https://kavrakilab.org/
	Graduate Student <i>Advisor: Dr. Lydia Kavraki</i>	Aug. 2016–Dec. 2021
	Undergraduate Researcher <i>Advisor: Dr. Lydia Kavraki</i>	Feb. 2015–Aug. 2016
Dexterous Robotics Lab <i>at NASA JSC, Houston, TX</i>	NSTRF Fellow <i>Supervisor: Dr. Julia Badger</i>	Aug. 2017–Aug. 2021 https://er.jsc.nasa.gov/er4/
	USRA Intern <i>Supervisor: Dr. Julia Badger</i>	Summer 2017
	Guest Researcher <i>Supervisor: Dr. Julia Badger</i>	Summer 2016
Multi-Robot Systems Lab <i>at Rice University, Houston, TX</i>	Undergraduate Researcher <i>Advisor: Dr. James McLurkin</i>	May 2014–May 2015 http://mrsll.rice.edu/

Education

Rice University Houston, TX	Ph.D. in Computer Science Thesis: <i>Toward Efficient and General Multi-Modal Planning</i> Advisor: Dr. Lydia E. Kavraki	Aug. 2016–Dec. 2021
	M.S. in Computer Science Thesis: <i>A Unifying Framework for Constrained Sampling-Based Planning</i> Advisor: Dr. Lydia E. Kavraki	Aug. 2016–Dec. 2017
	B.S. in Computer Science	Aug. 2012–May 2016



Publications

All publications are available on [my lab's website](#) and [my Google Scholar](#). Author names are ordered by contribution unless otherwise marked. Undergraduate students I advised are marked with ^U. Graduate students I advised are marked with ^G. Purdue students are marked in [gold](#). Publications after starting at Purdue have their label marked in [gold](#). [†] marks equal contribution.

Journal Articles

- J1.** Thai Duong, Clayton W. Ramsey^G, **Zachary Kingston**, Wil Thomason, and Lydia E. Kavraki. Ultrafast sampling-based kinodynamic planning via differential flatness. *IEEE Transactions on Robotics*, 2026. URL <https://arxiv.org/abs/2603.16059>. To Appear
- J2.** Weihang Guo^G, Theodoros Tyrovouzis, Emiliano Flores, Clayton W. Ramsey^G, **Zachary Kingston**, Ioan A. Şucan, Mark Moll, and Lydia E. Kavraki. The Open Motion Planning Library 2.0. *IEEE Robotics and Automation Magazine*, 2026. URL <https://arxiv.org/abs/2605.29301>. To Appear
- J3.** Qingxi Meng^G, Emiliano Flores, Carlos Quintero-Peña, Peizhu Qian, **Zachary Kingston**, Shan-nan K. Hamlin, Vaibhav Unhelkar, and Lydia E. Kavraki. Look as you leap: Planning simultaneous motion and perception for high-DoF robots. *IEEE Transactions on Robotics*, 2026. doi:[10.1109/TRO.2026.3731487](https://doi.org/10.1109/TRO.2026.3731487). URL <https://arxiv.org/abs/2509.19610>. To Appear
- J4.** Yuanchu Liang, Edward Kim, J. Arden Knoll, Wil Thomason, **Zachary Kingston**, Lydia E. Kavraki, and Hanna Kurniawati. Think fast and far: Long-horizon online POMDP planning via rapid state sampling. *The International Journal of Robotics Research*, 2026. doi:[10.1177/02783649261453546](https://doi.org/10.1177/02783649261453546)
- J5.** Francesco Fuentes, Serigne Diagne, **Zachary Kingston**, and Laura H. Blumenschein. Linking exteroception and proprioception through improved contact modeling for soft growing robots. *The International Journal of Robotics Research*, 2026. doi:[10.1177/02783649261448230](https://doi.org/10.1177/02783649261448230)
- J6.** Yitian Gao^{†G}, Lucas Chen^{†G}, Priyanka Bhovad, Sicheng Wang, **Zachary Kingston**, and Laura H. Blumenschein. Parallel simulation of contact and actuation for soft growing robots. *Soft Robotics*, 13(3):405–420, 2026. doi:[10.1177/21695172261425906](https://doi.org/10.1177/21695172261425906)
- J7.** Weihang Guo^G, **Zachary Kingston**, Kaiyu Hang, and Lydia E. Kavraki. Efficient multi-robot motion planning for manifold-constrained manipulators by randomized scheduling and informed path generation. *IEEE Robotics and Automation Letters*, 11(4):4385–4392, 2026. doi:[10.1109/LRA.2026.3662639](https://doi.org/10.1109/LRA.2026.3662639)

- J8.** S. Talha Bukhari^G, Kaivalya Agrawal^U, **Zachary Kingston**, and Aniket Bera. Variational shape inference for grasp diffusion on SE(3). *IEEE Robotics and Automation Letters*, 11(2):2170–2177, 2026. doi:[10.1109/LRA.2025.3645521](https://doi.org/10.1109/LRA.2025.3645521)
- J9.** Tyler S. Wilson, Wil Thomason, **Zachary Kingston**, and Jonathan D. Gammell. AORRTC: Almost-surely asymptotically optimal planning with RRT-connect. *IEEE Robotics and Automation Letters*, 10(12):13375–13382, 2025. doi:[10.1109/LRA.2025.3615522](https://doi.org/10.1109/LRA.2025.3615522)
- J10.** S. Bora Bayraktar, Andreas Orthey, **Zachary Kingston**, Marc Toussaint, and Lydia E. Kavraki. Solving rearrangement puzzles using path defragmentation in factored state spaces. *IEEE Robotics and Automation Letters*, 8(8):4529–4536, 2023. doi:[10.1109/LRA.2023.3282788](https://doi.org/10.1109/LRA.2023.3282788). Presented at ICRA 2024
- J11.** **Zachary Kingston** and Lydia E. Kavraki. Scaling multi-modal planning: Using experience and informing discrete search. *IEEE Transactions on Robotics*, 39(1):128–146, 2023. doi:[10.1109/TRO.2022.3197080](https://doi.org/10.1109/TRO.2022.3197080)
- J12.** Constantinos Chamzas, Carlos Quintero-Peña, **Zachary Kingston**, Andreas Orthey, Daniel Rakita, Michael Gleicher, Marc Toussaint, and Lydia E. Kavraki. MotionBenchMaker: A tool to generate and benchmark motion planning datasets. *IEEE Robotics and Automation Letters*, 7(2):882–889, 2021. doi:[10.1109/LRA.2021.3133603](https://doi.org/10.1109/LRA.2021.3133603)
- J13.** **Zachary Kingston**, Mark Moll, and Lydia E. Kavraki. Exploring implicit spaces for constrained sampling-based planning. *The International Journal of Robotics Research*, 38(10–11):1151–1178, 2019. doi:[10.1177/0278364919868530](https://doi.org/10.1177/0278364919868530)
- J14.** Neil T. Dantam, **Zachary Kingston**, Swarat Chaudhuri, and Lydia E. Kavraki. An incremental constraint-based framework for task and motion planning. *The International Journal of Robotics Research*, 37(10):1134–1151, 2018. doi:[10.1177/0278364918761570](https://doi.org/10.1177/0278364918761570)
- J15.** **Zachary Kingston**, Mark Moll, and Lydia E. Kavraki. Sampling-based methods for motion planning with constraints. *Annual Review of Control, Robotics, and Autonomous Systems*, 1(1):159–185, 2018. doi:[10.1146/annurev-control-060117-105226](https://doi.org/10.1146/annurev-control-060117-105226)

Refereed Conference Papers

- C1.** Shrutheesh R. Iyer^G, I-Chia Chang, Andrew Z. Liu^U, Yan Gu, and **Zachary Kingston**. Vectorizing projection in manifold-constrained motion planning for real-time whole-body control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2026. URL <https://arxiv.org/abs/2604.13323>. To Appear
- C2.** Qingyi Chen^G, **Zachary Kingston**, and Ahmed H. Qureshi. NeHMO: Neural Hamilton-Jacobi reachability learning for decentralized safe multi-agent motion planning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2026. URL <https://arxiv.org/abs/2607.00326>. To Appear
- C3.** Yitian Gao^G, Andrew Lu^U, and **Zachary Kingston**. Fast asymptotically optimal kinodynamic planning via vectorization. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2026. URL <https://arxiv.org/abs/2607.03987>. To Appear
- C4.** Cael Yasutake^U, Andrew H. Liu^U, **Zachary Kingston**, and Brian Plancher. HJCD-IK: GPU-accelerated inverse kinematics through batched hybrid jacobian coordinate descent. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2026. URL <https://arxiv.org/abs/2510.07514>. To Appear

- C5.** Le Mao, Andrew H. Liu^U, Renos Zabounidis, Yanan Niu, **Zachary Kingston**, and Joseph Campbell. CDE: Concept-driven exploration for reinforcement learning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2026. URL <https://arxiv.org/abs/2510.08851>. To Appear
- C6.** José Johil Patiño Miñán, **Zachary Kingston**, Victor Romero-Cano, Yu-Kun Lai, and Juan D. Hernández. One-shot view planning and online optimization-based replanning for unknown object reconstruction. In *IEEE International Conference on Robotics and Automation*, 2026. To Appear
- C7.** Chih H. Huang^{U†}, Pranav Jadhav^{†U}, Brian Plancher, and **Zachary Kingston**. pRRTC: GPU-parallel RRT-connect for fast, consistent, and low-cost motion planning. In *IEEE International Conference on Robotics and Automation*, 2026. URL <https://arxiv.org/abs/2503.06757>. To Appear
- C8.** Hanlan Yang^G, Itamar Mishani, Luca Pivetti, **Zachary Kingston**, and Maxim Likhachev. Parallel heuristic search as inference for actor-critic reinforcement learning models. In *IEEE International Conference on Robotics and Automation*, 2026. URL <https://arxiv.org/abs/2509.25402>. To Appear
- C9.** Lucas Chen^G, Shrutheesh R. Iyer^G, and **Zachary Kingston**. Differentiable particle optimization for fast sequential manipulation. In *IEEE International Conference on Robotics and Automation*, 2026. URL <https://arxiv.org/abs/2510.07674>. To Appear
- C10.** Mitchell E. C. Sabbadini, Andrew H. Liu^U, Joseph Ruan^U, Tyler S. Wilson, **Zachary Kingston**, and Jonathan D. Gammell. Revisiting replanning from scratch: Real-time incremental planning with fast almost-surely asymptotically optimal planners. In *IEEE International Conference on Robotics and Automation*, 2026. URL <https://arxiv.org/abs/2510.21074>. To Appear
- C11.** Akshaya Agrawal^G, Evan Palmer, **Zachary Kingston**, and Geoffrey A. Hollinger. Underwater multi-robot simulation and motion planning in angler. In *IEEE/MTS OCEANS Conference*, pages 1–6, Brest, France, 2025. doi:[10.1109/OCEANS58557.2025.11104649](https://doi.org/10.1109/OCEANS58557.2025.11104649)
- C12.** Lucas Chen^{†U}, Yitian Gao^{†U}, Sicheng Wang, Francesco Fuentes, Laura H. Blumenschein, and **Zachary Kingston**. Physics-grounded differentiable simulation for soft growing robots. In *IEEE-RAS International Conference on Soft Robotics*, pages 1–8, 2025. doi:[10.1109/RoboSoft63089.2025.11020809](https://doi.org/10.1109/RoboSoft63089.2025.11020809)
- C13.** Tyler S. Wilson, Wil Thomason, **Zachary Kingston**, Lydia E. Kavraki, and Jonathan D. Gammell. Nearest-neighbourless asymptotically optimal motion planning with fully connected informed trees (FCIT*). In *IEEE International Conference on Robotics and Automation*, pages 14140–14146, 2025. doi:[10.1109/ICRA55743.2025.11127785](https://doi.org/10.1109/ICRA55743.2025.11127785)
- C14.** Akshaya Agrawal^G, Parker Mayer, **Zachary Kingston**, and Geoffrey A. Hollinger. Constrained nonlinear Kaczmarz projection on intersections of manifolds for coordinated multi-robot mobile manipulation. In *IEEE International Conference on Robotics and Automation*, pages 7726–7732, 2025. doi:[10.1109/ICRA55743.2025.11127991](https://doi.org/10.1109/ICRA55743.2025.11127991)
- C15.** Weihang Guo^G, **Zachary Kingston**, and Lydia E. Kavraki. CaStL: Constraints as specifications through LLM translation for long-horizon task and motion planning. In *IEEE International Conference on Robotics and Automation*, pages 11957–11964, 2025. doi:[10.1109/ICRA55743.2025.11127555](https://doi.org/10.1109/ICRA55743.2025.11127555)
- C16.** Yuanchu Liang[†], Edward Kim[†], Wil Thomason[†], **Zachary Kingston**[†], Hanna Kurniawati, and Lydia E. Kavraki. Scaling long-horizon online POMDP planning via rapid state space sampling.

In *International Symposium of Robotics Research*, 2024. URL <https://arxiv.org/abs/2411.07032>

- C17. Clayton W. Ramsey^G, **Zachary Kingston**[†], Wil Thomason[†], and Lydia E. Kavraki. Collision-affording point trees: SIMD-amenable nearest neighbors for fast collision checking. In *Robotics: Science and Systems*, 2024. doi:[10.15607/RSS.2024.XX.038](https://doi.org/10.15607/RSS.2024.XX.038)
- C18. Wil Thomason[†], **Zachary Kingston**[†], and Lydia E. Kavraki. Motions in microseconds via vectorized sampling-based planning. In *IEEE International Conference on Robotics and Automation*, pages 8749–8756, 2024. doi:[10.1109/ICRA57147.2024.10611190](https://doi.org/10.1109/ICRA57147.2024.10611190)
- C19. Carlos Quintero-Peña, Wil Thomason, **Zachary Kingston**, Anastasios Kyrillidis, and Lydia E. Kavraki. Stochastic implicit neural signed distance functions for safe motion planning under sensing uncertainty. In *IEEE International Conference on Robotics and Automation*, pages 2360–2367, 2024. doi:[10.1109/ICRA57147.2024.10610773](https://doi.org/10.1109/ICRA57147.2024.10610773)
- C20. Khen Elimelech, **Zachary Kingston**, Wil Thomason, Moshe Vardi, and Lydia E. Kavraki. Accelerating long-horizon planning with affordance-directed dynamic grounding of abstract strategies. In *IEEE International Conference on Robotics and Automation*, pages 12688–12695, 2024. doi:[10.1109/ICRA57147.2024.10610486](https://doi.org/10.1109/ICRA57147.2024.10610486)
- C21. Rahul Shome, **Zachary Kingston**, and Lydia E. Kavraki. Robots as AI double agents: Privacy in motion planning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 2861–2868, 2023. doi:[10.1109/IROS55552.2023.10341460](https://doi.org/10.1109/IROS55552.2023.10341460)
- C22. Carlos Quintero-Peña, **Zachary Kingston**, Tianyang Pan, Rahul Shome, Anastasios Kyrillidis, and Lydia E. Kavraki. Optimal grasps and placements for task and motion planning in clutter. In *IEEE International Conference on Robotics and Automation*, pages 3707–3713, 2023. doi:[10.1109/ICRA48891.2023.10161455](https://doi.org/10.1109/ICRA48891.2023.10161455)
- C23. Yiyuan Lee^G, Wil Thomason, **Zachary Kingston**, and Lydia E. Kavraki. Object reconfiguration with simulation-derived feasible actions. In *IEEE International Conference on Robotics and Automation*, pages 8104–8111, 2023. doi:[10.1109/ICRA48891.2023.10160377](https://doi.org/10.1109/ICRA48891.2023.10160377)
- C24. **Zachary Kingston** and Lydia E. Kavraki. Robowflex: Robot motion planning with *MoveIt* made easy. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 3108–3114, 2022. doi:[10.1109/IROS47612.2022.9981698](https://doi.org/10.1109/IROS47612.2022.9981698)
- C25. **Zachary Kingston**, Constantinos Chamzas, and Lydia E. Kavraki. Using experience to improve constrained planning on foliations for multi-modal problems. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 6922–6927, 2021. doi:[10.1109/IROS51168.2021.9636236](https://doi.org/10.1109/IROS51168.2021.9636236)
- C26. Mark Moll, Constantinos Chamzas, **Zachary Kingston**, and Lydia E. Kavraki. HyperPlan: A framework for motion planning algorithm selection and parameter optimization. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 2511–2518, 2021. doi:[10.1109/IROS51168.2021.9636651](https://doi.org/10.1109/IROS51168.2021.9636651)
- C27. Andrew M. Wells, **Zachary Kingston**, Morteza Lahijanian, Lydia E. Kavraki, and Moshe Y. Vardi. Finite horizon synthesis for probabilistic manipulation domains. *IEEE International Conference on Robotics and Automation*, pages 6336–6342, 2021. doi:[10.1109/ICRA48506.2021.9561297](https://doi.org/10.1109/ICRA48506.2021.9561297)
- C28. Constantinos Chamzas, **Zachary Kingston**, Carlos Quintero-Peña, Anshumali Shrivastava, and Lydia E. Kavraki. Learning sampling distributions using local 3D workspace decompositions

- for motion planning in high dimensions. In *IEEE International Conference on Robotics and Automation*, pages 1283–1289, 2021. doi:[10.1109/ICRA48506.2021.9561104](https://doi.org/10.1109/ICRA48506.2021.9561104)
- C29. **Zachary Kingston**, Andrew M. Wells, Mark Moll, and Lydia E. Kavraki. Informing multi-modal planning with synergistic discrete leads. In *IEEE International Conference on Robotics and Automation*, pages 3199–3205, 2020. doi:[10.1109/ICRA40945.2020.9197545](https://doi.org/10.1109/ICRA40945.2020.9197545)
- C30. **Zachary Kingston**, Mark Moll, and Lydia E. Kavraki. Decoupling constraints from sampling-based planners. In Nancy M. Amato, Greg Hager, Shawna Thomas, and Miguel Torres-Torriti, editors, *Robotics Research*, pages 913–928. Springer International Publishing, Cham, 2020. ISBN 978-3-030-28619-4. doi:[10.1007/978-3-030-28619-4_62](https://doi.org/10.1007/978-3-030-28619-4_62)
- C31. Golnaz Habibi, Sándor P. Fekete, **Zachary Kingston**, and James McLurkin. Distributed object characterization with local sensing by a multi-robot system. In Roderich Groß, Andreas Kolling, Spring Berman, Emilio Frazzoli, Alcherio Martinoli, Fumitoshi Matsuno, and Melvin Gauci, editors, *Distributed Autonomous Robotic Systems: The 13th International Symposium*, volume 6, pages 205–218. Springer Proceedings in Advanced Robotics, 2018. doi:[10.1007/978-3-319-73008-0_15](https://doi.org/10.1007/978-3-319-73008-0_15)
- C32. William Baker, **Zachary Kingston**, Mark Moll, Julia Badger, and Lydia E. Kavraki. Robonaut 2 and you: Specifying and executing complex operations. In *IEEE Workshop on Advanced Robotics and its Social Impacts*, pages 1–8, Austin, TX, March 2017. doi:[10.1109/ARSO.2017.8025204](https://doi.org/10.1109/ARSO.2017.8025204)
- C33. Neil T. Dantam, **Zachary Kingston**, Swarat Chaudhuri, and Lydia E. Kavraki. Incremental task and motion planning: A constraint-based approach. In *Robotics: Science and Systems*, Ann Arbor, MI, June 2016. doi:[10.15607/RSS.2016.XII.002](https://doi.org/10.15607/RSS.2016.XII.002)
- C34. **Zachary Kingston**, Neil T. Dantam, and Lydia E. Kavraki. Kinematically constrained workspace control via linear optimization. In *IEEE-RAS International Conference on Humanoid Robots*, pages 758–764, Nov 2015. doi:[10.1109/HUMANOIDS.2015.7363455](https://doi.org/10.1109/HUMANOIDS.2015.7363455)
- C35. Golnaz Habibi, **Zachary Kingston**, Zijian Wang, Mac Schwager, and James McLurkin. Pipelined consensus for global state estimation in multi-agent systems. In *Proceedings of the 2015 International Conference on Autonomous Agents and Multiagent Systems*, pages 1315–1323. International Foundation for Autonomous Agents and Multiagent Systems, 2015. ISBN 9781450334136. doi:[10.5555/2772879.2773320](https://doi.org/10.5555/2772879.2773320)
- C36. Golnaz Habibi, **Zachary Kingston**, William Xie, Mathew Jellins, and James McLurkin. Distributed centroid estimation and motion controllers for collective transport by multi-robot systems. In *IEEE International Conference on Robotics and Automation*, pages 1282–1288, 2015. doi:[10.1109/ICRA.2015.7139356](https://doi.org/10.1109/ICRA.2015.7139356)

Book Chapters

- B1. **Zachary Kingston**. *Encyclopedia of Robotics*, chapter Planning Under Manifold Constraints, pages 1–9. Springer Berlin Heidelberg, 2020. ISBN 978-3-642-41610-1. doi:[10.1007/978-3-642-41610-1_174-1](https://doi.org/10.1007/978-3-642-41610-1_174-1)

Preprints

- P1. Servet B. Bayraktar, Andreas Orthey, **Zachary Kingston**, and Marc Toussaint. PEEL: Parallel extraction for long-horizon disassembly planning via scale-invariant sampling, 2026. URL <https://arxiv.org/abs/2608.08773>. Under Review

- P2.** S. Talha Bukhari^G, Yi Wei^U, Ruiqi Ni, **Zachary Kingston**, and Aniket Bera. Fast generative grasping via Lie group-constrained MeanFlow, 2026. URL <https://arxiv.org/abs/2608.26076>. Under Review
- P3.** Sai Coumar^G, An T. Le, and **Zachary Kingston**. Anytime global tensor motion planning, 2026. URL <https://arxiv.org/abs/2608.25830>. Under Review
- P4.** Pengyuan Guo, Zhonghao Mai, Zhengtong Xu, Kaidi Zhang, Heng Zhang, Zichen Miao, Arash Ajoudani, **Zachary Kingston**, Qiang Qiu, and Yu She. PPlanAR: Planning-language-grounded agentic reasoning for robot manipulation, 2026. URL <https://arxiv.org/abs/2602.01662>
- P5.** Muyang Yan^{†U}, Miras Mengdibayev[†], Ardon Floros^U, Weihang Guo^G, Lydia E. Kavraki, and **Zachary Kingston**. Using VLM reasoning to constrain task and motion planning, 2025. URL <https://arxiv.org/abs/2510.25548>. Under Review
- P6.** Sai Coumar^G, Gilbert Chang^U, Nihar Kodkani^U, and **Zachary Kingston**. Foam: A tool for spherical approximation of robot geometry, 2025. URL <https://arxiv.org/abs/2503.13704>
- P7.** Sai Coumar^U and **Zachary Kingston**. Evaluating machine learning approaches for ASCII art generation, 2025. URL <https://arxiv.org/abs/2503.14375>

Refereed Workshop Papers and Abstracts

- W1.** Itamar Mishani, Hanlan Yang^G, Luca Pivetti, **Zachary Kingston**, and Maxim Likhachev. Parallel heuristic search as inference for actor-critic reinforcement learning models (extended abstract). In *International Symposium on Combinatorial Search*, volume 19, pages 310–311, 2026. doi:[10.1609/socs.v19i1.43111](https://doi.org/10.1609/socs.v19i1.43111)
- W2.** Chih H. Huang^U, **Zachary Kingston**, and Brian Plancher. pAORRTC: GPU-parallel almost-surely asymptotically optimal planning. In *IEEE ICRA 2026 Workshop—RoboARCH: Robotics Acceleration with Computing Hardware and Systems*, 2026
- W3.** Sai Coumar^G, Weihang Guo^G, and **Zachary Kingston**. PyRoFFI: Accelerating foreign kinematics kernels. In *IEEE ICRA 2026 Workshop—RoboARCH: Robotics Acceleration with Computing Hardware and Systems*, 2026
- W4.** Alexiy Buynitsky^U and **Zachary Kingston**. Faster behavior cloning with hardware-accelerated motion planning. In *IEEE ICRA 2025 Workshop—RoboARCH: Robotics Acceleration with Computing Hardware and Systems*, 2025
- W5.** Qingxi Meng^G, Carlos Quintero-Peña, **Zachary Kingston**, Vaibhav Unhelkar, and Lydia E. Kavraki. Perception-aware planning for robotics: Challenges and opportunities. In *40th Anniversary of the IEEE Conference on Robotics and Automation (ICRA@40)*, 2024
- W6.** Carlos Quintero-Peña, Wil Thomason, **Zachary Kingston**, Anastasios Kyrillidis, and Lydia E. Kavraki. Stochastic implicit neural signed distance functions for safe motion planning under sensing uncertainty. In *IEEE ICRA 2024 Workshop—Back to the Future: Robot Learning Going Probabilistic*, 2024
- W7.** Clayton W. Ramsey^G, **Zachary Kingston**[†], Wil Thomason[†], and Lydia E. Kavraki. Dynamic motion planning from perception via accelerated point cloud collision checking. In *IEEE ICRA 2024 Workshop—Agile Robotics: From Perception to Dynamic Action*, 2024
- W8.** Qingxi Meng^{†G}, Carlos Quintero-Peña[†], **Zachary Kingston**, Nicole M. Fontenot, Shannan K. Hamlin, Vaibhav Unhelkar, and Lydia E. Kavraki. Monitoring constraints for robotic tutors in nurse education: A motion planning perspective. In *IEEE ICRA 2024 Workshop—Workshop on Nursing Robotics*, 2024

Theses

- T1. **Zachary Kingston.** *Toward Efficient and General Multi-Modal Planning.* PhD thesis, Rice University, Houston, TX, 2021. URL <https://hdl.handle.net/1911/111679>
- T2. **Zachary Kingston.** A unifying framework for constrained sampling-based planning. Master's thesis, Rice University, Houston, TX, 2017. URL <https://hdl.handle.net/1911/105607>

Invited Contributor

- I1. Claire Le Goues, Sebastian Elbaum, David Anthony, Z. Berkay Celik, Mauricio Castillo-Effen, Nikolaus Correll, Pooyan Jamshidi, Morgan Quigley, Trenton Tabor, and Qi Zhu. Software engineering for robotics: Future research directions; report from the 2023 workshop on software engineering for robotics, 2024. URL <https://arxiv.org/pdf/2401.12317.pdf>



Funding

Funding obtained after starting at Purdue has the label marked in *gold*.

Authored

- G1.** **Zachary Kingston.** Industrial grant withheld for confidentiality, 2026. \$179,876
- G2.** **Zachary Kingston** et al. Industrial grant withheld for confidentiality, 2025. \$203,062
- G3.** Laura Blumenschein and **Zachary Kingston.** Collaborative research: Building a collaborative network of researchers in mechano-computation, 2025. URL https://www.nsf.gov/awardsearch/showAward?AWD_ID=2527267. NSF CMMI 2527267, \$177,000
- G4.** **Zachary Kingston.** Industrial grant withheld for confidentiality, 2025. \$136,010
- G5. Rahul Shome, Jenny L. Davis, **Zachary Kingston**, and Lydia E. Kavraki. Virtues of Robot Inaction: Towards Theories of Automated Reasoning of Inaction in Human Contexts, 2023. ANU Humanising Machine Intelligence Computing for Social Good Seed Research Grant, AU\$25,000

Co-Authored

- cG1. Lydia E. Kavraki. Elements: Scalable Next-Generation Software Infrastructure for High-Dimensional Search, 2024–2027. URL https://www.nsf.gov/awardsearch/showAward?AWD_ID=2411219. NSF OAC 2411219, \$599,302
- cG2. Lydia E. Kavraki and Anshumali Shrivastava. A Framework for Manipulation Planning and Execution under Uncertainty in Partially-Known Environments, 2024–2027. URL https://www.nsf.gov/awardsearch/showAward?AWD_ID=2336612. NSF CCF 2336612, \$715,312
- cG3. Lydia E. Kavraki and Vaibhav Unhelkar. Collaborative Research: FW-HTF-R: The Future of Robot-Assisted Nursing: Interactive AI Frameworks for Upskilling Nurses and Customizing Robot Assistance, 2022–2023. URL https://nsf.gov/awardsearch/showAward?AWD_ID=2222876. NSF HRD 2222876, \$121,713
- cG4. Lydia E. Kavraki. RI: Small: A Novel Framework for Informed Manipulation Planning, 2020–2023. URL https://nsf.gov/awardsearch/showAward?AWD_ID=2008720. NSF IIS 2008720, \$441,000

Awards and Honors

- A1. **Rice Innovation Fellow**, Liu Idea Lab for Innovation and Entrepreneurship, 2024
- A2. **C24**. Nominated, Best Paper in Industrial Robotics Research for Practicality, IEEE/RSJ IROS, 2022
- A3. **C28**. Nominated, Best Paper in Cognitive Robotics, IEEE-RAS ICRA, 2021
- A4. **Future Faculty Fellowship**, Rice Engineering, 2020–21
- A5. Best Presentation in COMP 600, Rice University Computer Science Department, 2018, 2020
- A6. **NASA Space Technology Research Fellowship**, NASA, 2017–2021
- A7. **NSF Graduate Research Fellowship Program**, NSF, 2017
- A8. **Graduate Research Fellowship**, Rice University Computer Science Department, 2016
- A9. **Distinction in Research and Creative Works**, Rice University, 2016
- A10. **President’s Honor Roll**, Rice University, 2015–16



Research Supervision

Students who I am the primary advisor for have their label marked in **gold**.

Ph.D. Students

- D1**. Yitian Gao, Purdue University, 2026–Present
- D2**. Qingyi Chen, Purdue University, 2025–Present
- D3**. **Shruthesh R. Iyer**, Purdue University, 2025–Present
- D4**. **Sai Coumar**, Purdue University, 2025–Present
- D5**. **Hanlan Yang**, Purdue University, 2025–Present
- D6**. **Austin Garrett**, Purdue University, 2025–Present
- D7. Kevin Zhang, Purdue University, 2025–Present, with Lin Tan
- D8. **S. Talha Bukhari**, Purdue University, 2024–Present, with Aniket Bera
- D9. **Akshaya Agrawal**, Oregon State University, 2024–Present, with Geoffrey Hollinger
- D10. **Clayton Ramsey**¹, Rice University, 2023–2025, with Lydia Kavraki
- D11. **Weihang Guo**, Rice University, 2023–2025, with Lydia Kavraki
- D12. Qingxi Meng, Rice University, 2023–2025, with Lydia Kavraki
- D13. **Yiyuan Lee**, Rice University, 2022–2023, with Lydia Kavraki

¹ Awarded NSTGRO and NDSEG 2024

Master’s Students

- M1**. Kaivalya Agrawal, Purdue University, 2026–Present
- M2**. Andrew Niu, Purdue University, 2026–Present
- M3**. Lucas Chen, Purdue University, 2024–2025
- M4**. Yitian Gao, Purdue University, 2024–2025
- M5**. Alexiy Buynitsky, Purdue University, 2024–2025
- M6. Elliot Herring, Rice University, 2021–2022

Undergraduate Students

- U1.** Derek McTume, Purdue University, 2026–Present
- U2.** Yi Wei, Purdue University, 2026–Present
- U3.** Jolie Lai, Purdue University, 2026–Present
- U4.** Philip Wisniewski, Purdue University, 2026–Present
- U5.** Joseph Ruan, Purdue University, 2025–Present
- U6.** Gilbert Chang, Purdue University, 2025–Present
- U7.** Nihar Kodkani, Purdue University, 2025–Present
- U8.** Ardon Floros, Purdue University, 2025–Present
- U9.** Zihan Lin, Purdue University, 2025–Present
- U10.** Pranav Jadhav, Purdue University, 2024–Present
- U11.** Andrew Z. Liu, Purdue University, 2025–2026
- U12.** Andrew H. Liu, Purdue University, 2025–2026
- U13.** Muyang Yan, Purdue University, 2025–2026
- U14.** Kaivalya Agrawal, Purdue University, 2025–2026
- U15.** Andrew Niu, Purdue University, 2025–2026
- U16.** Andrew Lu, Purdue University, 2024–2026, CS 497 Project
- U17.** Richard Li, Purdue University, 2025, CS 497 Project
- U18.** Isaac Fuksman, Purdue University, 2025, CS 497 Project
- U19.** Aaron Boes, Purdue University, 2025, CS 497 Project
- U20.** Lucas Chen, Purdue University, 2024
- U21.** Yitian Gao, Purdue University, 2024
- U22.** Stefan Bukorovic, University of Belgrade, 2023
- U23.** Sofia Paola Medina-Chica, Pontificia Universidad Javeriana, 2023
- U24.** Andreja Andrejic, University of Belgrade, 2022
- U25.** Aedan Cullen, Rice University, 2022
- U26.** Luis Leal², Rice University, 2021

² Award for Excellence in Poster Presentations, Rice Summer Undergraduate Research Symposium

Visiting Scholars

- V1.** Miras Mengdibayev, Purdue University, 2025
- V2.** Sai Coumar, Purdue University, 2024–2025

Academic Committees

- AC1.** Xingpeng Sun, CS Ph.D., Advisor: Aniket Bera, Purdue University
- AC2.** Ruiqi Ni, CS Ph.D., Advisor: Ahmed Qureshi, Purdue University
 - ✦ Ph.D. Preliminary Exam, May 2025
 - ✦ Ph.D. Defense, Mar. 2026
- AC3.** Phu An Pham, CS Ph.D., Advisor: Aniket Bera, Purdue University
 - ✦ Ph.D. Preliminary Exam, Feb. 2025
 - ✦ Ph.D. Defense, Sep. 2026
- AC4.** Dipam Patel, CS Ph.D., Advisor: Aniket Bera, Purdue University
- AC5.** Maheep Brar, CS MS, Advisor: Joseph Campbell, Purdue University
- AC6.** Akshaya Agrawal, CS Ph.D., Advisor: Geoffrey Hollinger, Oregon State University
 - ✦ Ph.D. Preliminary Exam, Nov. 2025
- AC7.** Hrishikesh Viswanath, CS Ph.D., Advisor: Aniket Bera, Purdue University
- AC8.** Apoorva Vashisth, CS Ph.D., Advisor: Aniket Bera, Purdue University

- AC9. Maxwell Asselmeier, Robotics Ph.D., Advisor: Ye Zhao and Patricio A. Vela, Georgia Institute of Technology
 ✦ Ph.D. Preliminary Exam, May 2026
- AC10. Daniel Castro Mesa, CS Ph.D., Advisor: Jan-Anders E. Mansson and David F. Gleich, Purdue University
- AC11. Sophia Knap, ME Ph.D., Advisor: Laura H. Blumenschein, Purdue University
- AC12. Zhengyuan Li, CS Ph.D., Advisor: Raymond A. Yeh and Aniket Bera, Purdue University
- AC13. Haomeng Zhang, CS Ph.D., Advisor: Raymond A. Yeh, Purdue University
- AC14. Zixing Wang, CS Ph.D., Advisor: Ahmed Qureshi, Purdue University



Service

Program Committee & Associate Editor

- ✦ IEEE Robotics and Automation Letters (Planning and Simulation) (RA-L), 2024–Present
- ✦ Robotics: Science and Systems (RSS), 2026
- ✦ World Symposium on the Algorithmic Foundations of Robotics (WAFR), 2026
- ✦ IEEE International Conference on Robotics and Automation (ICRA), 2024–2027
- ✦ IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026
- ✦ Conference on Robot Learning (CoRL), 2026
- ✦ AAAI Conference on Artificial Intelligence (AAAI), 2026–2027
- ✦ IEEE International Conference on Simulation, Modeling, and Programming for Autonomous Robots (SIMPRA), 2025

Organizer

- ✦ **MESH Network Kick-off Workshop**, CMU, Pittsburgh, PA, 2026
- ✦ **RoboARCH: Robotics Acceleration with Computing Hardware and Systems**, ICRA 2025–2026
- ✦ **Evaluating Motion Planning Performance Workshop**, IROS 2022

Session Chair

- ✦ IEEE International Conference on Robotics and Automation (ICRA), 2025

Conference and Journal Referee Work

- ✦ IEEE International Conference on Robotics and Automation (ICRA), 2018, 2022–2026
- ✦ IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020–2026
- ✦ Robotics: Science and Systems (RSS), 2024–2025
- ✦ IEEE Robotics and Automation Letters (RA-L), 2020–2025
- ✦ IEEE-RAS International Conference on Soft Robotics (RoboSoft), 2025
- ✦ IEEE Transactions on Robotics (T-RO), 2022–2023, 2025
- ✦ Journal of Artificial Intelligence Research (JAIR), 2024
- ✦ IEEE/ASME Transactions on Mechatronics, 2020
- ✦ IEEE Transactions on Automation Science and Engineering (T-ASE), 2020
- ✦ Workshop on the Algorithmic Foundations of Robotics (WAFR), 2020
- ✦ International Symposium on Robotics Research (ISRR), 2017

Government Activities

- ✦ Reviewer for NSF CISE, 2025
- ✦ Reviewer and panelist for NSF CS Grad4US, 2025
- ✦ Reviewer and panelist for NSF CISE/ENG, 2025
- ✦ Proposed topic for NSF Emerging Frontiers in Research and Innovation (EFRI), reached Blue Ribbon Panel, 2025
- ✦ Reviewer for the NASA Space Technology Graduate Research Opportunity (NSTGRO), 2025

Department Service

- ✦ Graduate Admissions Committee, AY 2024–2025, AY 2025–2026

Outreach

- ✦ **Research Presentation and Demo for “Brick by Brick: Bringing AI to the Classroom”**, Jul. 2026
- ✦ Research Presentation and Demo for Thorntown Elementary, Thorntown, IN, Dec. 2025
- ✦ Research Presentation and Demo for Nextech students, West Lafayette, IN, Nov. 2025
- ✦ Research Presentation and Demo for Northfield Jr./Sr. Highschool, Wabash, IN, Jun. 2025
- ✦ Judge for Purdue SIGBots Slam and Jam VEXU Qualifier, Mar. 2025
- ✦ Consultant for Robot Revolution Exhibit at the Museum of Science and Industry, Chicago, IL, Jan. 2015–May 2015
- ✦ Summer Swarm Robotics Camp Instructor at Rice University, Jul. 2014

Other Service

- ✦ Maintainer of OMPL Motion Planning Software, Dec. 2021–Present
- ✦ Core Contributor to the *MoveIt* Robot Motion Planning Software, Mar. 2019–Jun. 2020
- ✦ Maintainer of the *MoveIt* Robot Motion Planning Software, Sep. 2018–Mar. 2019
- ✦ Computer Science Representative, Graduate Student Association, Mar. 2018–May 2020
- ✦ Consultant for Rice’s **Center for Academic and Professional Communication**, Aug. 2018–May 2019
- ✦ Treasurer, Computer Science Graduate Student Association, Aug. 2017–May 2019



Teaching

- T1.* **Data Science Capstone** (Fall 2026), Instructor of Record, cs 441-DSC, Purdue University, 21 Enrolled
- T2.* **Software Engineering for Robotics** (Spring 2026), Instructor of Record, cs 593-SER, Purdue University, 26 Enrolled
- T3.* **Data Science Capstone** (Fall 2025), Instructor of Record, cs 441-DSC, Purdue University, 60 Enrolled
- T4.* **Software Engineering for Robotics** (Spring 2025), Instructor of Record, cs 593-SER, Purdue University, 24 Enrolled
- T5.* **Robot Manipulation** (Fall 2024), Instructor of Record, cs 592-RM1, Purdue University, 5 Enrolled
- T6.* **Introduction to Robotics** (Fall 2024), Guest Lecturer, cs 458, Purdue University
- T7.* **Physical Computing** (Spring 2024), Instructor of Record, COMP 650, Rice University, 11 Enrolled
- T8.* **Algorithmic Robotics** (Fall 2020), Lecturer, COMP/ELEC/MECH 450/550, Rice University
- T9.* **Algorithmic Robotics** (Fall 2016–2019, 2022), Teaching Assistant, COMP/ELEC/MECH 450/550, Rice University

- T10. **Introduction to Computer Systems** (Spring 2015, 2018), In-Lab Teaching Assistant, COMP 321, Rice University
- T11. **Introduction to Computational Thinking** (Fall 2015), In-Class Teaching Assistant, COMP 140, Rice University
- T12. **Introduction to Engineering Systems** (Fall 2014), In-Class Teaching Assistant, ENGI 128, Rice University



Invited Talks

Talks are organized by talk title. If available, a link to a video of the talk is provided with the link.

Pathways into Robotics

Brick by Brick: Bringing AI to the Classroom at West Lafayette, IN Aug. 2026

Real-Time Motion Planning for Any Robot

Midwest Robotics Workshop 2026 at Atlanta, GA Jun. 2026

Speaking Pl{ai,a}nly: Language to Constraints and Back Again

AIDA3 Research Meeting at West Lafayette, IN Apr. 2026

Purdue CS Undergraduate Symposium at West Lafayette, IN Oct. 2025

Workshop on Language and Semantics of TAMP at ICRA, Atlanta, GA May 2025

Efficient and General Planning for Robot Manipulation

Purdue CS Graduate Symposium at West Lafayette, IN Mar. 2025

Purdue CS Graduate Seminar at West Lafayette, IN Oct. 2024

Purdue CS Honor's Seminar at West Lafayette, IN Sep. 2024

UCSD ECE Seminar at La Jolla, CA Mar. 2024

Purdue Computer Science Seminar at West Lafayette, IN Feb. 2024

Texas A&M Computer Science Seminar at College Station, TX Feb. 2024

Motions in Microseconds via Vectorized Sampling-Based Planning

Intelligent Autonomous Systems Group virtual at TU Darmstadt, w/ Wil Thomason Jan. 2024

Pumps & Pipes at the ION, w/ Wil Thomason Dec. 2023

Robonaut 2 and You: Specifying and Executing Complex Operations

International Workshop on AI-Powered Space at the ION Nov. 2023

Humanoid Users Conference at NASA JSC May 2017

Scaling Multi-Modal Planning

AI, ML, and Friends Seminar virtual at ANU School of Computing Feb. 2023

University of Utah Robotics Seminar virtual Jul. 2021

Learning and Intelligent Systems Lab virtual at TU Berlin Jul. 2021

Robowflex: Simplifying Planning and Benchmarking with MoveIt

2022 MoveIt Community Meeting virtual Feb. 2022

Open Source Software

Maintainer

The Open Motion Planning Library (OMPL)

<https://ompl.kavrakilab.org/>

Vector-Accelerated Motion Planner (VAMP)

<https://github.com/KavrakiLab/vamp>

Tracing Compiler for VAMP

<https://github.com/CoMMALab/cricket>

Differentiable Vine Simulator

<https://github.com/CoMMALab/DiffVineSimPy>

Foam

<https://github.com/CoMMALab/foam>

GPU-Parallel RRT-Connect

<https://github.com/CoMMALab/pRRTC>

Robowflex

<https://github.com/KavrakiLab/robowflex>

MotionBenchMaker

https://github.com/KavrakiLab/motion_bench_maker

HyperPlan

<https://github.com/KavrakiLab/hyperplan>

Contributor

MoveIt Robot Motion Planning Software

<https://moveit.ros.org/>